
AGE ESTIMATION WITH UNCERTAINTY

PROJECT REPORT

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December 4, 2025

ABSTRACT

Accurate age estimation from facial images is challenging due to large variation in appearance and noise in dataset labels. This project investigates whether a regression model can predict both age and uncertainty using a GoogLeNet backbone trained on the UTKFace dataset. We address distribution imbalance through stratified splitting and sampling techniques, and explore methods for improving model robustness, including data augmentation and ensembling. The project further analyzes the predicted uncertainty through confidence intervals and evaluates model interpretability using Grad-CAM techniques. Finally, we discuss the model's limitations and outline directions for improving uncertainty calibration in future work.

Keywords Deep Learning · Age Estimation · Transfer Learning · Regression · Confidence intervals

1 Introduction

Age estimation from facial images is a challenging and inherently noisy task. Individuals age differently due to factors such as genetics, lifestyle, and environment, and the provided labels may reflect either chronological age or apparent age, which introduces additional ambiguity. Traditional convolutional neural network models typically produce a single point estimate, which fails to capture the uncertainty that naturally arises from this variability.

In this project, we aim to build an age prediction model that accounts for this uncertainty. We apply transfer learning using the pre-trained GoogLeNet architecture [Szegedy et al., 2014], originally trained on ImageNet, and adapt its output layer to predict not only the expected age but also an associated predictive variance. This enables us to construct confidence intervals for each prediction and evaluate both the accuracy and reliability of the model's uncertainty estimates.

Our goal is to investigate whether incorporating uncertainty leads to more interpretable and robust predictions, and to assess how well these uncertainty estimates are calibrated. We further explore how model design choices and evaluation procedures influence the quality of both predictions and uncertainty estimates.

2 Related work

Prior research has emphasized the importance of consistent and transparent model evaluation, particularly in age related prediction tasks. [Paptham and Franc, 2023] point out the lack of standardized data splits and the simultaneous modification of multiple components can make it difficult to attribute improvements to specific model changes. They advocate for public data splits and controlled experiments that modify only one component at a time, ensuring that observed performance gains result from the model or training procedure rather than inconsistencies in benchmarking or data partitioning. Moreover, they highlight the importance of maintaining a consistent age distribution across *training*, *validation*, and *test* data splits. Such recommendations guides the approach of our experiments and ensure that observed performance differences stems from the model rather than data inconsistencies.

Proper evaluation of predictive uncertainties in regression tasks has been extensively studied. [Kuleshov et al., 2018] demonstrate that uncertainty estimates are often poorly calibrated, meaning that predicted confidence intervals do not align with empirical residuals. Building on this, [Levi et al., 2019] critique existing calibration definitions and propose metrics such as coverage and sharpness to better assess the quality of uncertainty estimates. These studies highlight the importance of evaluating both the accuracy of predictions and the reliability of the associated uncertainty. Following this line of work, our project employs a Gaussian Negative Log-Likelihood loss to predict both mean age and variance, which we then use to construct confidence intervals for evaluation.

The MiVOLO paper [Kuprashevich and Tolstykh, 2023] represents one of the current state of the art approaches in age and gender estimation, and is therefore a relevant baseline for comparing our results. Unlike our approach, which will rely solely on facial input, MiVOLO uses a multi-input transformer that processes both a face crop and a full-body crop. This fusion of features through cross attention improves robustness, particularly in unconstrained settings where the face may be partially occluded or visually ambiguous.

3 Methods

This section describes the structured approach used in the project, including dataset characteristics, preprocessing and splitting strategies, and the modeling approach for predicting both age and predictive uncertainty. Together, these components form the foundation for the experimental setup evaluated in later sections.

3.1 Data analysis

We use the UTKFace dataset [Song and Zhang], which contains over 20,000 facial images annotated with age, gender, and race. The age labels range from 1 to 116 years, offering broad coverage of all ages. The images have been cropped to only contain the face, this focus on facial regions makes the dataset well aligned with our task of age estimation from facial features. The gender distribution is 52.3% male and 47.7% female, ensuring minimal bias toward gender specific features. Race is also reasonably balanced relative to age, but Asians have a lower mean age (25.87) compared to Indians (31.53), Blacks (33.86), and Whites (37.98). However, the age distribution is highly imbalanced: 32.22% of samples fall within ages 0–25 and 48.35% within 26–50, while individuals above 75 represent less than 4% of the dataset.

This imbalance poses a challenge for regression models, as underrepresented age groups, particularly older individuals may lead to systematic bias and increased prediction uncertainty. The images also vary considerably in lighting, contrast, and overall quality. Such inconsistencies can obscure age related facial cues and make the regression task more difficult, potentially increasing residual variance and reducing model robustness.

3.2 Data split

The dataset is divided into training (70%), validation (20%), and test (10%) subsets. To maintain demographic representativeness, we use stratified sampling based on coarse 20-year age intervals combined with gender. This ensures that the relative proportions of age and gender groups remain consistent across all splits.

The partitioning is fully deterministic, guaranteeing that the same samples appear in the same subsets across all experiments. This design is essential for reproducibility and allows for fair comparison between different model configurations.

Figure 1 shows the resulting age distributions. The raw counts confirm that the training subset contains the majority of samples across all age groups, while the validation and test subsets mirror the same overall profile at smaller scale. The normalized view further demonstrates that age proportions are preserved across splits. Notably, the density of samples decreases sharply after age 90, reflecting the dataset’s natural imbalance.

Analyzing the datasets age distribution across train, validation and test helps us spot and address potential issues. Figure 1a shows that the age distribution is non-uniform, with data mainly concentrated around the age of 20-35, and with a spike at the very young ages around 0. Figure 1b shows that the train, validation and test splits has a similar age distribution. It also shows that the data for ages above 90 becomes more sparse. This non-uniform distribution may introduce bias, particularly for underrepresented older age groups.

3.3 Uncertainty and confidence

Our model predicts two values for each input image: a mean age prediction μ and a variance σ^2 . The mean represents the expected age, while the variance reflects the model’s uncertainty inherent to the data, such as noise in labels or

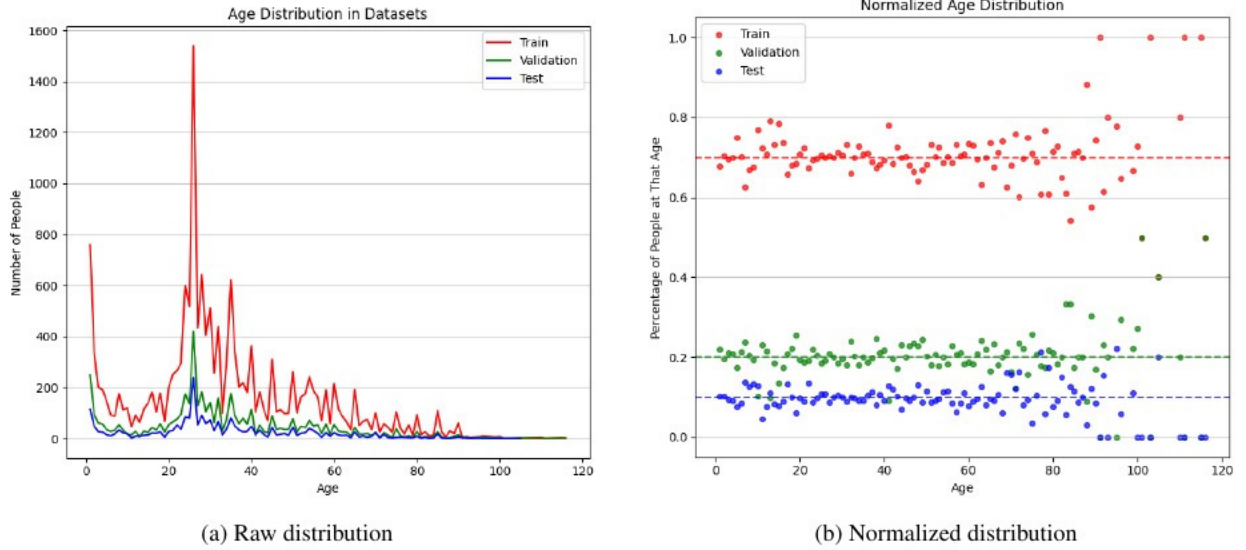


Figure 1: Comparison of raw and normalized data distributions.

ambiguous facial features. Both quantities are learned jointly by minimizing the Gaussian Negative Log-Likelihood (NLL) loss from Pytorch:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{2} \left(\log(\max(\sigma^2, \varepsilon)) + \frac{(y - \mu)^2}{\max(\sigma^2, \varepsilon)} \right) + \text{const.}$$

Here, y is the true age label, and ε is a small constant for numerical stability. This loss penalizes large residuals but also discourages the model from arbitrarily increasing σ^2 , because overestimating variance increases the logarithmic term. As a result, the model learns to output high variance only when predictions are genuinely uncertain.

After training the predicted, mean and variance can be used to calculate a confidence interval around the predicted mean:

$$\text{CI}_{1-\alpha} = \mu \pm z_{1-\alpha/2} \cdot \sigma$$

where $z_{1-\alpha/2}$ is the corresponding standard normal quantile. These intervals provide an interpretable measure of the model’s confidence and form the basis for evaluating calibration in later experiments.

3.4 Network Architecture

We adopt GoogLeNet as the backbone encoder for our transfer learning setting, following the procedure outlined in transfer learning scenario two from [Pedersen 2025a]. GoogLeNet expects RGB images of size 224×224, so all UTKFace images are preprocessed using `torchvision` transforms. Each image is resized to the required dimensions and normalized using the standard ImageNet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]), ensuring compatibility with the pre-trained weights.

The original fully connected classification layer is removed and replaced with a custom regression head. The output from the encoder is flattened into a one dimensional feature vector and passed through a sequence of `Linear` → `BatchNorm1d` → `LeakyReLU` blocks. The final layer produces a two dimensional output (μ, σ^2) , corresponding to the predicted mean age and the predicted variance. We employ `LeakyReLU` activations to mitigate the dying ReLU problem and `BatchNorm1d` to stabilize and accelerate training, allowing for higher learning rates and improved convergence properties [Pedersen 2025b].

Figure 2 shows how transfer learning is utilized. To reduce overfitting, we apply horizontal flipping as data augmentation to 50% of the images in the training set. Augmentation is applied exclusively to the training split; the validation and test sets remain unchanged to ensure fair evaluation. Across all experiments, we use a batch size of 64, which provided a good balance between stable gradient estimates, efficient GPU utilization, and effective batch normalization. The subsequent sections will outline several experiments carried out in an attempt to enhance our model.

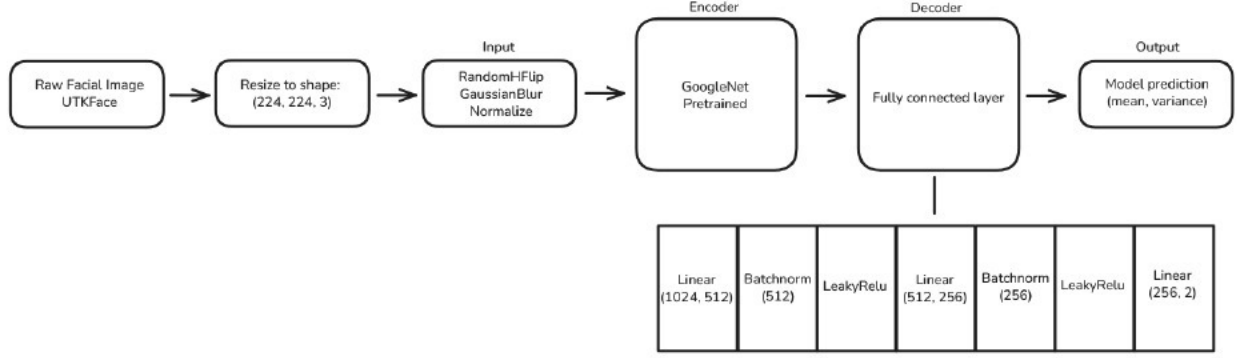


Figure 2: Model architecture

3.5 Regression experiments

We began with a simple fully connected regression network and gradually increased its capacity to better match the complexity of the UTKFace dataset. Our final baseline model, shown in Figure 3, uses GaussianNLLLoss and is optimized with stochastic gradient descent (momentum = 0.9, weight decay = $1 \cdot 10^{-4}$). We initially experimented with a deterministic learning rate schedule but ultimately adopted a plateau based scheduler that reduces the learning rate by a factor of $\gamma = 0.5$ when the validation loss stagnates.

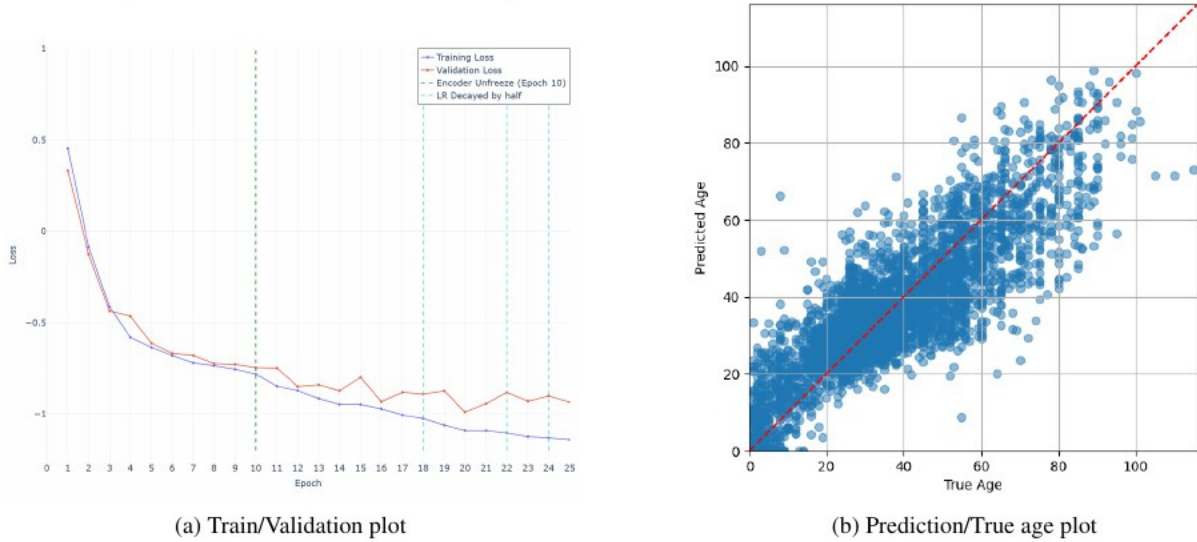


Figure 3: Regression model

The training loss decreases rapidly during the first eight epochs and begins to plateau around epoch 9 (Figure 3a). At epoch 10, we unfreeze the GoogLeNet encoder to increase model capacity, which initially improves performance but later introduces mild overfitting around epoch 14. The final baseline model achieves a test NLL of -1.0719 and a mean absolute error (MAE) of 6.59 years.

Figure 3b shows that predictions follow a clear linear trend but exhibit systematic bias: the model tends to overestimate ages below 20 and underestimate ages above 45. This bias is likely due to the dataset imbalance described in Section 3.2 where young and middle aged individuals are heavily overrepresented.

To mitigate bias toward frequent age groups, we incorporated PyTorch’s `WeightedRandomSampler`. Labels were grouped into decade based bins, and sampling weights were computed as $(\frac{1}{\alpha_x})^{0.3}$, where α_x denotes the proportion of samples in bin x . This increases the probability of selecting images from underrepresented age ranges during training.

The loss trajectory in Figure 4a closely resembles that of the baseline model, with rapid initial improvement followed by mild overfitting after the encoder is unfrozen. The sampler results in slightly more uniform predictions for ages 60–90 (Figure 4b), but has limited effect on ages above 90, where data remain sparse.

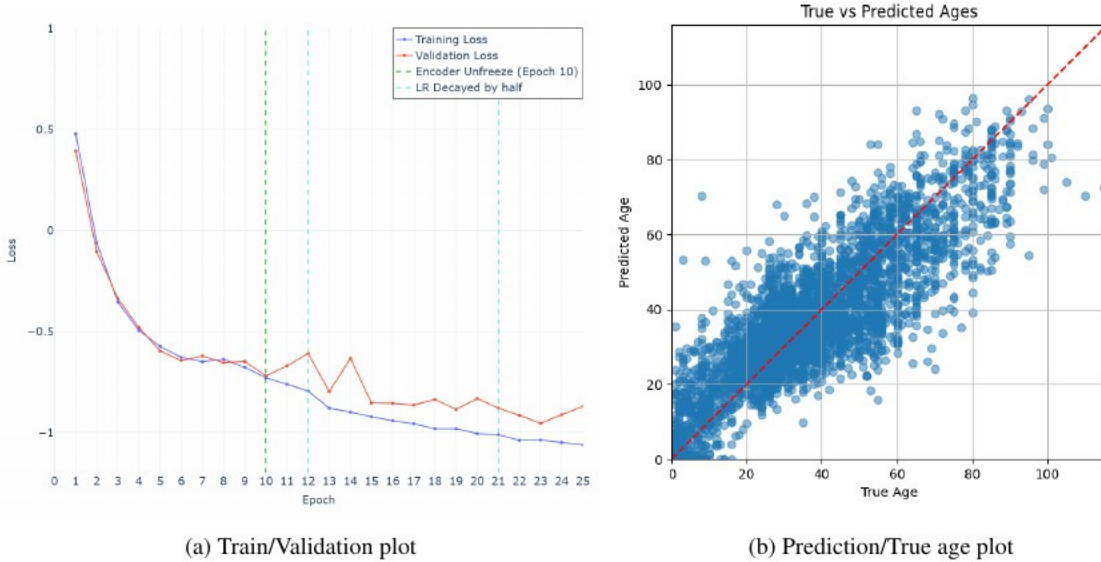


Figure 4: Regression model with WeightedRandomSampler

The sampler model achieves a test NLL of -1.0385 and MAE of 6.87 - slightly worse than the baseline. However, these aggregate metrics mask slight improvements within less represented portions of the dataset.

To quantitatively assess whether the *WeightedRandomSampler* mitigates age related bias, we compare the mean absolute error (MAE) per age decade for both the baseline and sampler augmented models, summarized in Table I

Table 1: Mean Absolute Error (MAE) per decade for models trained with and without the *WeightedRandomSampler*.

Model	0–9	10–19	20–29	30–39	40–49	50–59	60–69	70–79	80–89	90–99	100+
Baseline	6.35	8.88	5.87	4.69	7.68	10.63	10.86	10.84	17.34	16.06	31.90
WRS	5.33	7.70	5.15	5.85	8.73	10.87	10.11	9.74	13.69	13.51	31.74
Diff	-1.02	-1.18	-0.72	1.16	1.05	0.24	-0.75	-1.1	-3.65	-2.55	-0.16

While the overall test loss slightly worsens with the sampler (from -1.0719 to -1.0385), the decade wise results show that errors generally decrease for older and underrepresented age groups (0-29 and 60+), suggesting a modest reduction in bias even if the global performance metric remains unchanged.

To address variability in image quality, lighting, and contrast, we experimented with applying a *Gaussian blur* to the images during preprocessing, aiming to reduce high frequency noise and standardize feature representation. However, this approach produced negligible improvement if not a slight decline in model performance, suggesting that the model was either robust to these variations or that blurring removed subtle age related features critical for accurate prediction.

3.6 Variance experiments

To improve the stability and robustness of the model’s predictions, we explored an ensemble based approach. Neural networks trained independently on the same data often exhibit substantial variance due to random initialization, stochastic optimization, and sensitivity to specific training samples. In regression tasks, this variance directly affects both the predicted age and the predicted uncertainty. Ensembles mitigate these effects by averaging predictions across multiple models, reducing random fluctuations and improving reliability.

Ensemble Size	MAE	Avg. Variance
Base Model	6.59	161.430
2	6.562	146.245
3	6.508	145.244
4	6.532	150.384
5	6.558	151.966
6	6.525	149.854

Table 2: Performance of ensembles with different numbers of independently trained models.

Each ensemble member was trained with identical hyperparameters but different random seeds to promote diversity. During evaluation, each model produced both a mean age prediction and an associated predicted variance. The ensemble prediction was then obtained by averaging the models’ mean predictions and their predicted variances.

Choosing the appropriate ensemble size requires balancing potential performance gains against computational cost. Because our ensemble prediction is obtained by averaging the individual models’ mean and variance outputs, additional models help only when they provide complementary information—either through lower individual error or through uncorrelated prediction noise. In practice, this means that variance reduction tends to plateau once model diversity becomes limited.

To evaluate this effect, we tested ensembles of 2–6 independently trained models and compared them to the base model. The results are shown in Table 2. As expected, moving from one to multiple models yields moderate improvements: MAE decreases slightly and the average predicted variance drops from 161.43 to around 145–152. However, after the first few models, the gains diminish. A 3-model ensemble achieves the lowest MAE and the lowest average variance, while larger ensembles fluctuate slightly without providing consistent improvement.

Overall, a small ensemble of 2–3 models captures nearly all available improvements, both in accuracy and in predicted uncertainty. Larger ensembles offer only marginal changes, suggesting that the models are not diverse enough for additional members to yield substantial benefit.

To further reduce prediction variance, we introduced a weighted ensemble method where each model’s contribution depends on its predicted uncertainty. Instead of averaging all models equally, we assign weights proportional to the inverse of each model’s predicted variance, meaning the more confident models have greater influence on the final prediction.

The weights assigned to model i is calculated:

$$w_i = \frac{1/\sigma_i^2}{\sum_j 1/\sigma_j^2}$$

Using these weights, the ensemble prediction is computed as a weighted mean:

$$\mu_{ensemble} = \sum_i w_i \mu_i, \quad \sigma_{ensemble}^2 = \sum_i w_i \sigma_i^2$$

Ensemble Size	MAE	Avg. Variance
Base Model	6.59	161.430
2	6.555	142.435
3	6.500	140.415
4	6.527	143.679
5	6.554	144.639
6	6.516	142.226

Table 3: Performance of precision weighted ensembles with varying numbers of models.

Table 3 summarizes the results. Compared to uniform averaging, the weighted forward pass consistently reduces the average predicted variance and provides small but robust improvements in MAE. As before, the best overall performance is achieved with three models.

Overall, weighting by inverse variance yields modest but consistent benefits, while increasing the ensemble size beyond three models again provides only marginal gains.

4 Results

In this section, we evaluate our age estimation models trained on the UTK-Face dataset and compare them with previously published results. In addition to reporting point estimates, we also analyze the uncertainty calibration of the predicted age distributions.

4.1 Age estimation

To see how our models hold up, we compare our models against several established methods reported in the MiVOLO paper Kuprashevich and Tolstykh [2023]. The results are summarized in table 4.

Although our models does not match the performance described in the MiVOLO paper, this result is expected for several reasons. First, MiVOLO uses a multi-input transformer architecture that incorporates both face and body crops.

Model	MAE
CORAL	5.39
Randomized Bins	4.55
MWR	4.37
VOLO-D1 face	4.23
Our weighted ensemble model	6.50
Our ensemble model	6.508
Our baseline model	6.59
Ours balanced sampler	6.87

Table 4: Comparison with the MiVOLO paper

The use of ensembling improves our results modestly, reducing the MAE from 6.59 to 6.5 and the variance from 161 to ~ 140 . The improvement confirms that ensemble averaging helps reduce predictive variance. However, the ensemble still trails behind state-of-the-art models trained on larger and more diverse datasets.

4.2 Confidence bounds

In addition to point estimates, our model predicts a Gaussian distribution over ages for each input, allowing us to quantify uncertainty. For each test sample, we compute a 68% confidence interval, which gives the range in which the true value is expected to lie with 68% confidence, derived from the predicted mean and standard deviation. The evaluation of these intervals is summarized in Table 5.

Metric	Single GoogLeNet	Weighted ensemble (3 models)
Observed coverage (68% nominal)	83.6%	84.2%
Calibration gap	+15.6%	+16.2%
Mean interval width (years)	23.07	22.94

Table 5: Evaluation of predicted 68% confidence intervals on the UTKFace test set.

These results indicate that the model’s predicted uncertainty in age estimation is overconservative. For a nominal 68% confidence interval, the observed coverage is substantially higher (83–84%), yielding positive calibration gaps of over 15%. This means the predicted intervals are wider than necessary to achieve the nominal coverage. The mean interval width is roughly 23 years, with the ensemble providing a slight reduction compared to a single GoogLeNet model.

In practice, this suggests that while the model reasonably captures uncertainty in age predictions, it tends to overestimate the variability of ages. Consequently, the predicted intervals are conservative, often larger than needed.

5 Discussion

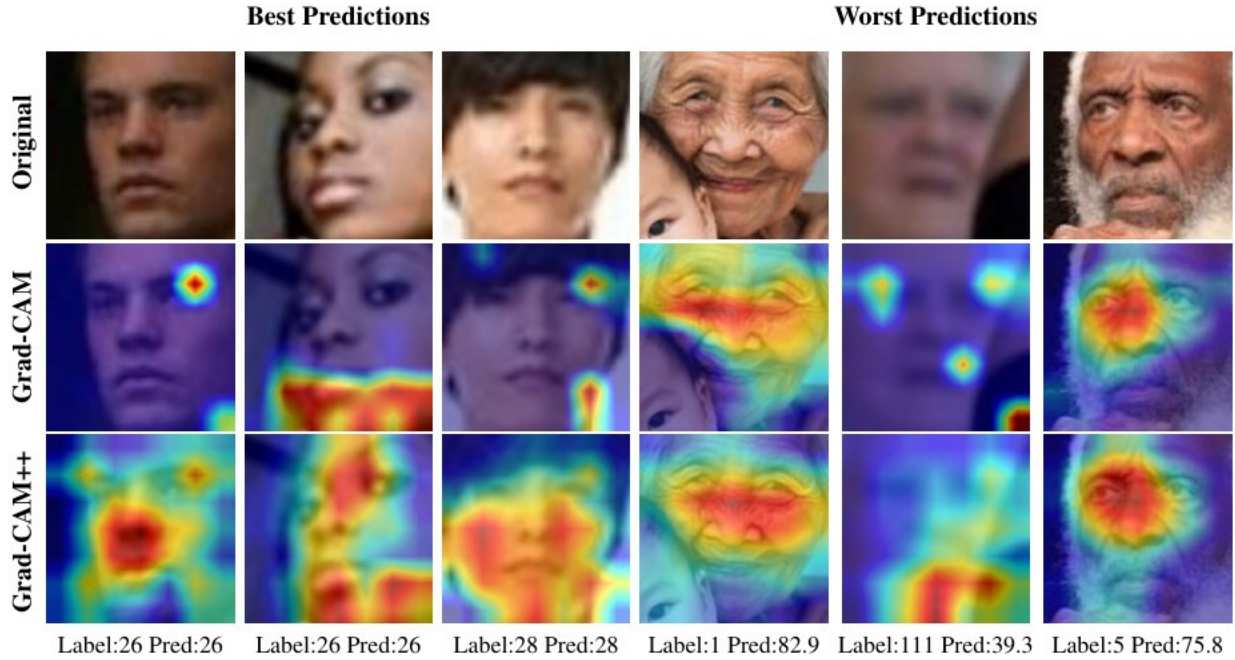
To better understand our model’s decision making, we visualized the most influential regions of the input images using both Grad-CAM and Grad-CAM++. These visualizations were generated using a single model trained without the balanced sampler. We selected six samples in total, three of the most accurate and the three of the least accurate predictions, and generated class activation heatmaps with the `pytorch_grad_cam` package [Gildenblat and contributors 2021] with activations taken from the last convolutional layer before global average pooling. This layer was chosen because it provides high level semantic features while maintaining spatial information necessary for meaningful heatmaps. The resulting heatmaps, shown in Figure 5, were overlaid on the original cropped facial images. Grad-cam seems to highlight more specific parts or points of the image than grad-cam++, but in general it looks like the model’s attention is predominantly centered on the core facial region, particularly around the nose, eye, and cheek area suggesting that these features are informative for age estimation.

However, the incorrectly predicted samples reveal important points of failure. Upon inspection, two out of the three worst predictions appear to be mislabeled, likely due to the dataset construction process, the images were originally part of larger images and the faces later cropped out, some were probably multi person photographs, and during face cropping, the visible face does not necessarily correspond to the annotated age. In one instance the main face is clearly elderly, yet the label corresponds to a baby that are also partly present in the cropped image. Such label noise can significantly distort training signals and lead to apparently large prediction errors, and these issues only became evident to us at the final stage of the project. Another point of failure is image quality. One of the worst predictions, labeled as age 111, is extremely blurry and hence lacks facial detail. The Grad-CAM and Grad-CAM++ visualizations for this image reflect this, highlighting mainly dark regions in the bottom right part of the image, indicating that the model could not effectively focus on meaningful facial features.

Our work partly focuses on evaluating whether ensemble modeling could improve both accuracy and uncertainty estimation over a single network. While the improvements were modest, the results show a consistent trend, ensembles help reduce the variance of predictions and improve robustness. The single GoogLeNet model achieved a MAE of 6.59, whereas the 3-model ensemble reduced this to 6.50, and also produced lower average predictive variance. This aligns with the expected behaviour of ensembles. By averaging multiple independent predictors, the influence of outlier predictions decreases and uncertainty estimates become more stable.

However, the gains diminish quickly beyond two or three models, suggesting limited model diversity. Since all models share the same architecture, dataset, and training procedure, their learned representations are likely similar. As a result, adding more models does not inject additional information into the ensemble, and accuracy plateaus. One possible extension would be to explore hybrid ensembles combining models trained with and without the balanced sampler, as

Figure 5: Original images, Grad-CAM and Grad-CAM++ visualizations for the 3 best and 3 worst predictions.



the two approaches learn from age distributions in different ways. This may introduce greater prediction diversity and provide a clearer understanding of how balanced training influences uncertainty and generalization.

The calibration results of the confidence bounds also highlight a challenge for future work, while our model is able to produce uncertainty estimates, the confidence intervals are overly conservative. Both the single model and the ensemble exhibit much higher coverage than expected, indicating that the predicted variance is too large relative to the empirical error. A natural next step for future work is to explore some uncertainty calibration methods. One method inspired by Levi et al. [2019] is to apply a post-hoc variance-scaling calibration. Rescaling the predicted standard deviations using a separate calibration set not used in training so that predicted uncertainty more closely matches empirical error. This could tighten confidence intervals and yield more informative uncertainty estimates.

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